

5.1 Canal Irrigation

- A canal is an artificial channel constructed on ground to carry water from a river or tank or reservoir for purposes like irrigation, power generation, navigation etc.
- Canal and channel means the same thing.
- Canals are usually having a trapezoidal cross section.
- Canals for the purpose of irrigation are usually open channels through earth or rock formation.

NOTE: Irrigation Canals can also be utilized for transportation of goods as well as inland navigation

5.2 Classification of Irrigation Canals

Irrigation canals can be classified in different ways on the following basis.

- (i) Based on nature of source of supply
 - (a) Permanent Canals
 - Perennial Canals
 - Non Perennial Canals
 - (b) Inundation Canals (or Flood Canals)
- (ii) Based on function of canals
 - (a) Feeder Canals
 - (b) Carrier Canals
- (iii) Based on discharge and relative importance in a given network of canals
 - (a) Main Canals
 - (b) Branch Canals
 - (c) Major distributary
 - (d) Minor distributary
 - (e) Water courses or Field channels
- (iv) Based on Canal alignment
 - (a) Ridge Canal
 - (b) Side Slope Canal
 - (c) Contour Canal
- (v) Based on soil through which it is constructed
 - (a) Alluvial Canals
 - (b) Non Alluvial Canals

- It carries full supply throughout the irrigation season

5.5 Layout of an Irrigation Canal Network

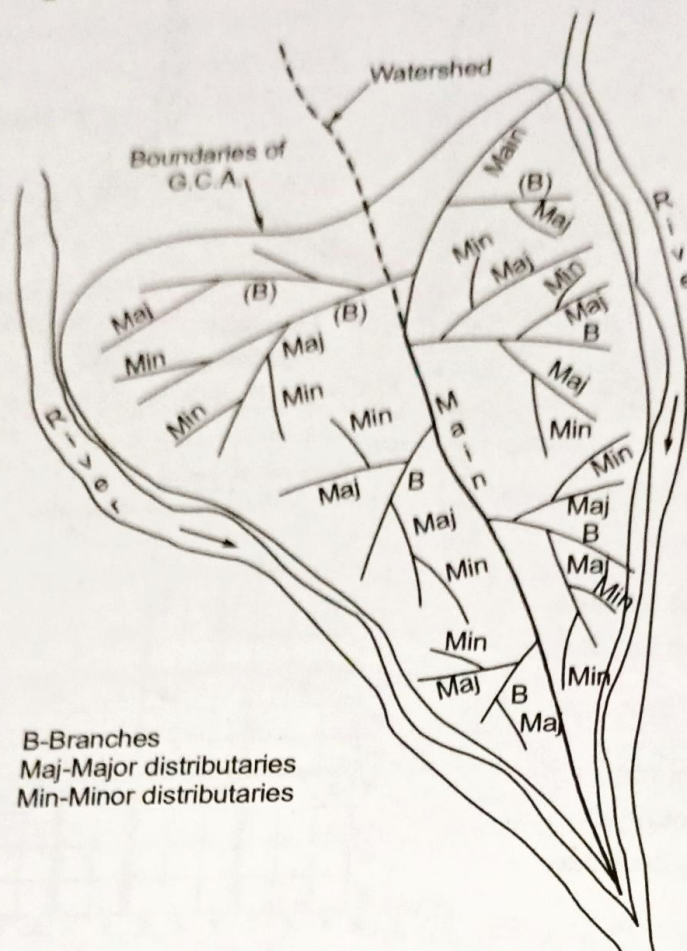


Fig. 5.3

5.5.1 Main Canal

- Principal canal of a canal network.
- Takes off at headworks directly from a river or reservoir.
- Direct irrigation is not usually carried out from the main canal. (Since a very high discharge is conveyed through the main canal)
- It is a large capacity canal (Supplies water to branch canals and major distributaries)

5.5.2 Branch Canal

- Branch Canals are the irrigation canals which take off from the main canal on either side.
- Large branches → no direct irrigation.
- Small branches → direct outlets are provided for irrigation of large areas.
- Generally carry a discharge of over 5 cumecs

§.3

Major Distributaries

- Major distributaries are irrigation canals which takes off from branch canals or sometimes from the main canal.
- Generally carry a discharge in the range between 0.25 to 5 cumecs
- They are generally used for direct irrigation

§.4

Minor Distributaries

- Minor distributaries are irrigation canals which take off from major distributaries and branch canals.
- Generally, carry a discharge less than 0.25 cumecs.

§.5

Water Courses (or field channels)

- Water courses are small channels which carries water from outlets of a major or minor distributary or a branch canal to the fields to be irrigated.
- Outlets are provided in the irrigation canals at required places.
- Beyond the outlet, water is handled by the individual cultivators who directs it to the various parts of the culturable command area.